

SAI DHIREN MUSALOJI

Agentic AI Engineer | MLOps & Systems Design

musalojidhiren@gmail.com | +1 862-423-8830 | [LinkedIn](#) | musalojidhiren.vercel.app

SUMMARY

Agentic AI Engineer with production experience spanning agentic AI, LLM infrastructure, and full-stack delivery. Designs and ships complete systems from backend to UI: API design (FastAPI, Node.js/Express, REST, GraphQL), data layers (PostgreSQL, Redis), agent orchestration and tool integration, and front-end interfaces (React, Next.js/SSR, TypeScript). Focused on closing the gap between AI research and reliable, cost-defensible production systems. Comfortable owning system design from scratch: local/private LLM inference and serving, cloud infrastructure (AWS, Azure, GCP), and containerized deployment (Docker, Kubernetes, Terraform).

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript, TypeScript

Frontend: React, Next.js (SSR), TypeScript, JavaScript, HTML, CSS, component-based/responsive UI development

Backend & APIs: FastAPI, Node.js/Express, RESTful API development & design, GraphQL, service orchestration, MCP-style tool gateways, WebSockets, Streamlit

AI & LLM Frameworks: LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen, DSPy, Model Context Protocol (MCP)

LLM Inference & Serving: vLLM, TensorRT-LLM, Triton Inference Server, Ollama, local/private model hosting, API gateway design

Vector Databases & Search: Pinecone, ChromaDB, FAISS, Weaviate, Qdrant, PGVector, Azure AI Search

ML & Data Tools: PyTorch, TensorFlow, Scikit-learn, XGBoost, PySpark, Pandas, NumPy, MLflow, Weights & Biases

Cloud & Infrastructure: AWS (S3, EC2, SQS, Rekognition, IAM, VPC), Azure AI Services, Google Cloud Platform, Docker, Kubernetes, Terraform (IaC)

Data & Backend Systems: PostgreSQL, Redis, Apache Spark, Hadoop/MapReduce, Apache Oozie, Apache Kafka, Databricks

Architecture & Practices: Component-based / modular architecture, event-driven & asynchronous design, distributed systems fundamentals, agent-to-agent communication and tool orchestration, responsive/accessible UI development (React, Next.js, Streamlit)

EXPERIENCE

AI Engineer | Tech Mahindra Americas

Mar 2026 – Present

- **Built a homegrown MCP-based agentic framework** with a tool suite (read, write, grep, OS-level sandboxed execution, and more) powering a production web UI (React, Next.js, TypeScript — customer auth, persistent chat history, file attachments) and a CLI agent, both able to scaffold and modify full projects autonomously.
- **Created Context Hub, a token-efficient knowledge-graph retrieval layer for coding agents:** indexed repositories once into a typed graph, code parsing for structure and dependency edges, project docs (BRDs, TDDs, OpenAPI specs) for meaning, instead of embedding and chunking everything into a vector store.
- **Engineered Context Hub's retrieval engine**, anchoring on BM25 and graph traversal to fetch only the relevant files per prompt, cutting tokens per query from ~5-15k (chunk-based RAG) to ~1.5-4k and 2-5 tool calls down to one; exposed it via an MCP server to agents like Cursor and Claude Desktop.
- **Deployed and served multiple LLMs locally** using vLLM, TensorRT-LLM, and Triton Inference Server on NVIDIA GPU hardware, then built an internal API gateway so other projects could consume local inference directly, keeping proprietary prompts and data off public APIs.
- **Architected a multi-agent conversational platform ("Refill")** on FastAPI and PostgreSQL: stage-aware billing/refill journeys, multi-agent orchestration, and a RESTful, MCP-style tool gateway (registered tools, invoke/audit path, conversation audit tables) with clear API contracts and full tool-call traceability.

AI/ML Engineer | Tawin Solutions LLC (USA)

Sept 2025 – Feb 2026

- **Designed a personalized recommendation engine** (collaborative + content-based filtering) incorporating user preference history, geolocation signals, and contextual features to improve suggestion relevance for an end-user-facing product.
- **Integrated Azure AI services with a structured logging/telemetry pipeline** to capture real-time user interactions and venue data, enabling continuous feedback loops for model retraining and drift detection.
- **Engineered feature pipelines from behavioral/event data**, translating raw interaction history into embeddings and structured signals consumed downstream by the recommendation service.

AI/ML Intern | Tech Mahindra, Makers Lab

Oct 2023 – Dec 2023

- **Contributed to Project Indus**, India's first multilingual LLM for native Indian languages; built data engineering pipelines (collection, cleaning, formatting of low-resource language corpora) and supported transformer architecture work during pre-training.
- **Built a real-time multilingual transcription service** for German-to-English meeting audio using Python-based ASR/NLP tooling, optimized for low-latency streaming processing and domain-specific vocabulary accuracy.

- **Provisioned and automated cloud infrastructure** across AWS (EC2, S3, IAM, VPC), Azure, and GCP using Terraform, writing reusable Infrastructure-as-Code configurations for repeatable environment setup across deployment targets.

PROJECTS

Intelligent Research Copilot: Full-Stack RAG Platform

- Developed an end-to-end research automation platform with a FastAPI backend exposing REST and GraphQL endpoints for ingestion, semantic retrieval, and abstractive summarization, paired with a React/Next.js (TypeScript) front-end, and a Streamlit variant for rapid internal iteration. Separated ingestion, retrieval, and presentation into independent, swappable layers for easier maintenance and scale.

Real-Time Ride Matching Engine: Distributed Backend System

- Engineered a low-latency rider-driver matching backend using Redis geospatial indexing, PostgreSQL state management, and WebSocket-based real-time updates for live client communication. Prototyped distributed, event-driven coordination and agent-to-agent assignment flows for real-time dispatch.

Event-Driven Vision Pipeline on AWS: Cloud-Native Backend Service

- Deployed a cloud-native detection pipeline integrating Amazon S3, Rekognition, and SQS behind an asynchronous service layer for large-batch processing. Added confidence-based filtering to cut false positives at scale.

Distributed Flight Analytics with Hadoop + Oozie

- Implemented large-scale flight performance analytics using Hadoop MapReduce, orchestrating multi-stage batch jobs with Apache Oozie to deliver delay/on-time trend analysis across high-volume datasets.

City-Scale Traffic Forecasting: Spatiotemporal ML Service

- Developed a spatiotemporal forecasting system (LSTM/ARIMA) for short-horizon traffic speed prediction on sensor data, with preprocessing, training, and evaluation pipelines built as reusable components for intelligent transportation applications.

EDUCATION

Master of Science in Data Science, New Jersey Institute of Technology

Bachelor of Technology in Electronics & Communications, Mahatma Gandhi Institute of Technology

CERTIFICATIONS

- Azure AI Engineer Associate (Microsoft AI-102)
- Azure Data Scientist Associate (Microsoft DP-100)
- Databricks Generative AI Fundamentals (Databricks)